

# Visualising Collision Spot Uncertainty with Augmented Reality

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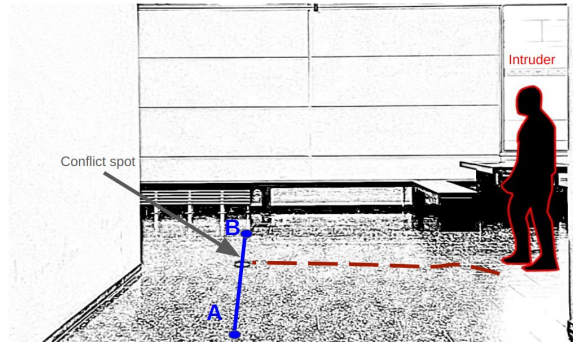


Figure 1: The figure shows a collision spot along a virtual path AB of an ego user and the path of another pedestrian.

## ABSTRACT

When trying to avoid collision in a scene with other persons, one makes an initial guess of a location where the conflict might happen before choosing paths. Then the estimated conflict spot serves as a deciding factor in choosing a safe trajectory to avoid the intruder. However, the predicted collision point is subject to one's estimate of the other person's future movement. The prediction is linked to guess made with a degree of uncertainty. In this paper, we summarize on how we use AR visualizations to symbolize different levels of uncertainty of predicted collision spots to study path choice influences

**Index Terms:** collision avoidance, walking behaviours, interaction, path prediction

## 1 INTRODUCTION

While walking in either indoor or outdoor scenes, the ease and comfort to walk past other pedestrians or cyclists is often based on how a person is able to predict potential conflicts along their chosen paths. If the walking scene is characterised by higher uncertainty due to shared motion with other road users (like cars, cyclists, etc.), then safety threats demand higher situational awareness and hence greater scene motion prediction accuracy. Shared spaces [9] are urban street designs that demand caution while also encouraging pedestrians to interact with cars and cyclists. Walking persons share movement space, communicate crossing intentions (using eye contact, hand signals or gestures) and negotiate priority, while expecting to maintain adequate distances to avoid blocking other road users. Due to a higher requirement to stay attentive, pedestrians have expressed safety concerns [19], which have led to criticism of the adaption of mixed traffic street designs.

Augmented Reality (AR) with its capability to superimpose 3D content into the scene has found itself useful in the field of navigation and wayfinding ([7], [5]). Some recent works have used AR as a safety medium to indicate crossing intentions ([1], [4]) as pedestrians interact with other road users. However, lesser works have focused on understanding the navigation effect of seeing AR content during safety-critical pedestrian interactions. While subconsciously prompting new routes to the destination has been the focus of AR manipulation techniques ([3, 10]), research on walked trajectories that would result when seeing AR content during pedestrian collision avoidance is interesting. This is primarily because such conflict avoidances are characterised by uncertainty due to randomness of human behaviour, or caused due to the AR content that is shown. For instance, Kamalasanan et al. [11] investigated whether showing the future walking path while crossing an intruder enhanced motion predictability and safety. The study results indicated towards safer crossing when confronting path intruders, while also identifying a correlation between chosen safe paths and their distances to the predicted point of collision. The participants in the study were repelled by the conflict point, choosing to walk around more to avoid collision with AR future paths. While visualisations applied to map navigation have nudged map users to choose longer routes to the destination [8], it is equally interesting to understand, if seeing virtual content for collision avoidance can prompt shorter and safer routes. Hence, in this work, we discuss our AR concept interface with the objective of studying AR content to prompt shorter avoidance routes. We also explain the procedure of our planned study on how the visualisation will be tested.

## 2 DESIGN OF UNCERTAINTY VISUALISATION

Uncertainty visualisation is an active research topic in the area of geographic information systems, scientific and information visualisation (e.g., [13], [2], [18]). As the idea of the visualisation is to impart information that may be inaccurate but also suggest its accuracy level, the information has to be communicated using different channels. Hence for our collision context (as in Figure 1), for a given time  $t$ , when indicating the collision spot, the visualisation should indicate spatially where a potential collision will happen ( $x_0$ ) and the level of certainty ( $\sigma$ ) with which an algorithm predicts the collision point C with the intruder when moving along the virtual

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ego path AB.

$$C_{AB}(t) = (x_0, \sigma) \quad (1)$$

While common channels of visual representation (like color) are occupied for showing the conflict position  $x_0$ , other techniques like the use of other visual variables [16] to represent uncertainty is used as an alternative approach in our work.

Visual variables are fundamental elements used to represent information graphically, facilitating the effective communication of data through visual means. The concept of visual variables was first introduced by Bertin [15], who identified eight primary variables: position (x,y), size, shape, color hue, color value, orientation, and texture. Furthermore, studies by Koussoulakou et al. in [12] revealed that animation enhances the understanding of messages compared to traditional static approaches. Consequently, DiBiase et al. [6] proposed dynamic visual variables, such as moment, frequency, duration, rate of change, order, and synchronization. Over time, visual variables have been further expanded and refined through additional techniques ([14, 20]). Inspired by recent works of applying static visual variables complemented by dynamics [17], we use the visual variable *texture* and apply *animations* and *size* variation to represent uncertainty in conflict spots. The spot is always predicted as a point on the straight line path of an ego pedestrian as s/he walks and crosses the intruder. Then a very uncertain collision spot (point with large  $\sigma$ ) would mean that the scene contains estimated collisions of low certainty and vice versa for certain collisions.

The uncertainty of the collision point can be expressed by the point error ellipse. Then the two radii along x and y axis define the uncertainty in estimated collision position for these two dimensions. Furthermore, there are primarily two ways to visualize this information. First by a pseudo-3D representation (isotropic) of a Gaussian distribution with a higher probability density in the middle and lower towards the boundary. Secondly, by showing individual sampled points from the same error distribution (anisotropic) simultaneously. Then the primary representation conveys a rather crisp and definite error boundary, while the secondary communicates the same but using a non deterministic or amorphous visualisation.

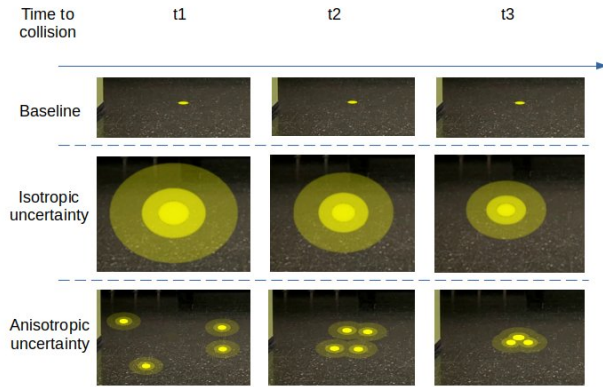


Figure 2: The figure shows the ego-users visualisation for *baseline* along with *isotropic* and *anisotropic* uncertainty AR for time instance  $t_1$ ,  $t_2$  and  $t_3$  where  $t_1 < t_2 < t_3$  seconds

Figure 2 illustrates the AR representation of our *baseline* (collision spot predicted as a crisp spot with no uncertainty information visualised) and the two designed variants for visualising conflict spot uncertainty, *isotropic* and *anisotropic* uncertainty AR for different time stamps  $t_1$ ,  $t_2$  and  $t_3$  before a collision ( $t_1 > t_2 > t_3$ ).

Our concept of a conflict uncertainty spot is represented using a yellow ellipse of concentric circles with decreasing radius (equivalent to increasing collision intensity) to its center. For the two variants *isotropic* and *anisotropic*, we apply the animations to the yellow ellipse differently. In the *isotropic* variant, we animate the radii( $r$ ) to depict how the algorithm confidence of a predicted collision spot will increase (indicated with reducing  $r$ ) or decrease (with increasing  $r$ ). We fix the radii ( $r = \text{constant}$ ) for the *anisotropic* version and represent uncertainty by animating motion of four yellow ellipses with positions relative to the center of conflict spot for time  $t$ . The quadruplet representation is further animated to move at boundary ellipse locations of the isotropic variant for the same time stamp. A key differentiation of anisotropic AR is its ability to show conflict positions on the floor in the presence of occlusions (like wall) due to ego view. Hence for a given error ellipse and scene boundary information, the visualisation represents uncertainty avoiding constraints from the environment.

The distance of the intruder from the conflict spot is used as a reference when applying animation. Then isotropic and anisotropic variants have larger radius and greater spatial separation between the quadruplets respectively at time  $t_1$  in comparison to time instance  $t_2$  (for increasing certainty). Both representations converge to a similar view beyond  $t_3$  seconds. To prototype the uncertainty designs, AR videos are created using post processed content creation. At first, raw video sequences are created using a static camera positioned in ego view, and single walking persons (intruder) in it. Then a virtual ego path AB that creates a collision with the person in the video clip is designed as in [11]. Animations and AR content are added based on the variant and intruder positions.

### 3 SCENES AND USER STUDY

As the objective of the work is to test AR visualisations for communicating collision spot uncertainty when an intruder crosses the ego user path, we create a sequence of ego-view videos that capture different levels of conflict point estimation difficulties for a intruder-ego path. We categorise the scenes of the sequence as:

- Increasing certainty: In these scenes, the intruder motion increases the certainty that a collision will be occurring at the estimated collision spot.
- Decreasing certainty: The certainty of a collision in these scenes shifts from a highly certain to an uncertain one. Such uncertainty results when the intruder decides to stop or change direction from an otherwise obvious conflict path with the virtual ego path.
- Consistent (un-)certainty: Walking patterns that result when the intruder is indulged in some secondary task (like reading etc.) causing zigzag motion patterns and consistent uncertain paths around the point of conflict. Then conflict points remain uncertain for such scenes.

For our work, we hypothesise that if an AR medium communicates the uncertainty of the collision point location, then the walked trajectories that result from avoiding conflicts with the intruder (for different scenes) would be shorter and/or longer paths around the conflict point. To test our hypothesis, we are interested in how participants as ego-users would move between the start and end points (A and B) of the virtual ego path while deciding to choose path deviations from AB to avoid collision with the intruder.

For this we plan to test our concept visualisation via a web based user study as in [11]. We first show AR content videos with either the baseline, isotropic or anisotropic variants of uncertainty. Having watched the video sequences, participants choose a walking path that they consider to be collision free, by sketching the path starting from A and ending in B. Then responses captured for

the baseline variant will help understand how route choices made when seeing the uncertain AR variants prompt different walking behaviours. While our current work mainly focuses on single intruder crossing scenes with a virtual ego path, our concept AR is also scalable to support multiple crossing situations.

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